

COMPLETE ANSWER KEY

DSAT_Mar_2025_INT_v4_W

All 147 Questions — Answers & Explanations

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SECTION 1: READING & WRITING — MODULE 1 (27 Questions)

Q1. Answer: A) detect

Why? The passage talks about realistic pieces where the viewer can easily see familiar objects. 'Detect' means to notice or spot something — it fits perfectly. 'Recall' means to remember (wrong idea). 'Dismiss' means to ignore (opposite). 'Classify' means to sort into groups (not about seeing).

Q2. Answer: B) appreciation of

Why? Critics helped people value radio by comparing it to concerts. 'Appreciation of' means having a positive view of something. 'Contribution to' doesn't make sense here. 'Imitation of' and 'dependence on' miss the point about building respect for the medium.

Q3. Answer: A) meticulousness

Why? The illustrators checked living plants, dried samples, AND expert scientists to make sure every detail was right. That is extreme care = meticulousness. 'Obscurity' means being unknown. 'Brevity' means shortness. 'Spontaneity' means being unplanned — all wrong.

Q4. Answer: A) settle in

Why? The sentence describes companies grouping together in one area. 'Settle in' means to set up in a place. 'Defer to' means to give in to someone. 'Compete with' and 'agree on' don't describe moving to a location.

Q5. Answer: C) familiar to

Why? The last sentence is the clue: fancy stage setups 'were not common until the 1800s.' So before that, simple stages were normal — audiences were used to them. 'Familiar to' fits. Simple stages wouldn't be 'puzzling' or 'thrilling' if they were the everyday standard.

Q6. Answer: A) undermine

Why? Bad data from inexperienced volunteers weakens the survey results. 'Undermine' means to damage or weaken. 'Confirm' is the opposite. 'Advertise' and 'predict' don't relate to weakening reliability.

Q7. Answer: D

Why? First the text says people think highly of Mr. Hargrove. Then it lists the things he DOES to keep that good image (stays quiet, hides his thoughts, never gives people a reason to doubt him). So: good reputation first, then behaviors that maintain it. That is choice D.

Q8. Answer: D

Why? The text describes a phenomenon (parallel adaptation), then mentions two studies that each looked at it differently (Ishida on shared pathways, Kokko on distinct pathways), then mentions a third study (Ramanathan and Delgado combining both). That matches D: phenomenon → two studies → third study.

Q9. Answer: B

Why? The narrator says she was 'never before so conscious of standing at a turning point.' She feels this specific moment is a big life change. That is B. She's not excited about money (A), she doesn't have a plan (C), and she doesn't mention thanking people (D).

Q10. Answer: A

Why? The energy staircase says: when people earn more money, they switch to cleaner fuels. So a supporter would explain India's change simply: incomes rose → families could afford cleaner fuel. That is A. Choice B talks about subsidies (a different explanation). C talks about biomass getting expensive (not the staircase model). D says the link weakened (contradicts the model).

Q11. Answer: A) 58,000 metric tons of tin in 2022

Why? The table shows Indonesia: 52 in 2000, 58 in 2022. The sentence says 'Indonesia produced 52 thousand in 2000 and _____. ' You just read the 2022 column for Indonesia = 58. Other choices use numbers from different countries.

Q12. Answer: A

Why? The claim: the quilts come from the maker's own life experience. Choice A says the child knows Grandma Rose 'Never found her patterns in any catalogue' — they came 'Straight from the soil of her own days.' This directly shows the quilts are born from personal experience.

Q13. Answer: C

Why? The passage says the gap between spoiled and unspoiled ratings VARIES across novels. You need two novels where that gap is DIFFERENT from each other. 'The Last Lantern' has a big gap (~1.0) and 'Ironside' has a small gap (~0.5). That contrast best shows the variation.

Q14. Answer: C

Why? The claim: smaller size is an adaptation to the prey in the MAB range. Choice C says MAB turtles feed at depths that bigger turtles CAN'T reach. This directly explains WHY being small is helpful — they can get food that large turtles cannot. That supports the adaptation idea.

Q15. Answer: A

Why? The passage warns: the novel is mostly imagined, so readers who try to find too many real-life connections are 'imposing unwarranted correspondences.' They're forcing connections that don't really exist. That is A. B is the opposite problem. C and D are off-topic.

Q16. Answer: B) from

Why? 'Species collected from rivers and coastal waters' — 'from' starts a prepositional phrase. No punctuation is needed after it. Adding a semicolon, colon, or comma would break the phrase apart incorrectly.

Q17. Answer: D) educate

Why? The subject is 'galleries' (plural). You need a present-tense verb that matches: 'galleries educate visitors.' The other options (having educated, educating, to educate) are not proper main verbs for the sentence.

Q18. Answer: D) it

Why? The pronoun refers back to 'the single "My Guy"' — one song, so you need a singular pronoun. 'It' is singular. 'Those' and 'they' are plural. 'Each' doesn't fit.

Q19. Answer: A) gain

Why? The sentence describes how things generally work (present tense). The subject 'employees' is plural. 'Gain' is present tense and plural — it matches. 'Had gained,' 'gained,' and 'were gaining' are all past tenses, which are wrong here.

Q20. Answer: D) century, revealing

Why? You need to connect 'twentieth century' with 'revealing how...' A comma + 'revealing' makes a smooth participial phrase. A period or semicolon before 'Revealing' would create an incomplete sentence. No punctuation at all creates a run-on.

Q21. Answer: D) On the other hand,

Why? The passage compares a sleek rowing shell (low drag) to a flat barge (high drag). These are opposites, so you need a contrast word. 'On the other hand' shows contrast. 'As a result' shows cause-effect. 'Specifically' adds detail. 'In summary' wraps up.

Q22. Answer: D) however,

Why? The passage says rings outside the Roche limit SHOULD become moons. But Quaoar's ring did NOT. This is unexpected. 'However' shows surprise/contrast. The other transitions don't capture this unexpected twist.

Q23. Answer: D

Why? Goal: emphasize the SIZE of Zacatecas. Only D mentions the actual area: 'approximately 75,040 km².' The other choices talk about the number of federal entities or public services — not size.

Q24. Answer: B

Why? Goal: show a CONTRAST between the two fungi. B says both are found on the Faroe Islands, but 'only the former is native.' One is native, one is not — that's a clear contrast. Other choices either don't compare them or don't highlight a difference.

Q25. Answer: C

Why? Goal: show WHEN each supercontinent formed relative to the other. C says 'Gondwana coalesced hundreds of millions of years after Rodinia.' That's a direct time comparison. A only mentions one. B and D don't compare their timing.

Q26. Answer: B

Why? Goal: show a SIMILARITY. B says 'Both silk fibers and nylon filaments lose tensile strength when subjected to cyclic loading.' Same thing happens to both = similarity. C highlights a difference (how MUCH they weaken). A describes the study. D is general.

Q27. Answer: D

Why? The audience already knows what germination is, so don't define it. D gives the specific findings (temperate woodland, acidic sandy soil, mulch helped germination) without explaining what germination means. B wastes time defining a term the audience already knows.

SECTION 1: READING & WRITING — MODULE 2 EASIER (27 Questions)

Q1. Answer: A) volatile

Why? After the colon, the passage describes a delta that is 'perpetually shifting.' The word must mean 'constantly changing.' 'Volatile' means subject to rapid change. 'Unchanging' is the opposite. 'Matchless' means unrivaled. 'Resilient' means tough — neither fits.

Q2. Answer: C) pioneering

Why? The tale is called 'the earliest recognizable work in the genre' — it was the first of its kind. 'Pioneering' means being a trailblazer. 'Conventional' means ordinary. 'Obscure' means unknown. 'Impartial' means neutral.

Q3. Answer: A) thoroughness

Why? The staff controlled humidity, lighting, AND temperature — very comprehensive care. 'Thoroughness' means completeness and attention to detail. The other words (mystery, simplicity, originality) don't describe how careful the preservation is.

Q4. Answer: C

Why? The text introduces two institutions and tells you what each one focuses on: one on Latino art/culture, the other on broad Latino contributions. That matches C. It doesn't discuss how they got their space (A), compare collection sizes (B), or trace a historical trend (D).

Q5. Answer: A

Why? The text first describes what all Oaxacan carvers share (same materials and motifs), then shows how one artist (Jimenez) added his own unique touch (finer details). Shared tradition → individual variation = choice A.

Q6. Answer: B

Why? Volkov says earlier models underestimated how much tidal energy the ice absorbs. So Tanaka's model was based on heating estimates that were too LOW (too conservative). Volkov would say: your heating numbers are too small, which puts the HZ boundary in the wrong place.

Q7. Answer: B

Why? The Edificio Garza Sada is described as having 'ornamental motifs and warm masonry' in a Neocolonial style. This CONTRASTS with his later Brutalist work. So it reflects an ornamental approach he later moved away from.

Q8. Answer: B

Why? The finding: seawalls hurt invertebrate communities MORE than revetments do. To show this, you need: a seawall-heavy site with LOW scores vs. a revetment-heavy site with HIGH scores. That is exactly what B describes (Drum Island seawalls < Shutes Folly revetments).

Q9. Answer: A

Why? The graph shows ALL plant types dropping in 2021. The ecologists say pickleweed dominance doesn't make the ecosystem tough. A explains both: a drought made salinity so extreme that even salt-tolerant plants suffered, proving the ecosystem isn't truly resilient despite pickleweed.

Q10. Answer: C

Why? The biologist says GPS gives LONGER distances than ARD. The bar-tailed godwit was measured by ARD at 4,600 km. If GPS had been used instead, it would record all detours and give a BIGGER number. C correctly says the godwit's distance 'would be greater than 4,600 km if GPS had been used.'

Q11. Answer: B

Why? The text says stable voltage is 'essential' for instruments. B supports this by saying 'irregular voltage compromises instrument accuracy.' If unstable power ruins accuracy, then stable power is indeed essential. That directly backs up the claim.

Q12. Answer: A

Why? The passage lists multiple unresolved obstacles and failed solutions. 'Consequently' introduces the logical conclusion: retailers probably can't fix these last-mile problems anytime soon. The other choices go beyond what the passage actually says.

Q13. Answer: C

Why? To find OLDER fossils, you need rocks from an OLDER time period. C says the Australian formation is from an earlier geological period than the Canadian one. That directly explains why researchers go to Australia for older specimens.

Q14. Answer: D

Why? Adding gold nanoparticles doubled the current. Normally, electrons have to go through slow, inefficient reactions. Gold conducts electricity well. So the hypothesis: gold lets electrons skip those slow reactions and go straight to the circuit. That is D.

Q15. Answer: C

Why? The model uses a FIXED CO2 value, but real CO2 probably VARIES. So when the model's predictions don't match reality, it could be because the real CO2 is different from the constant the model uses. C points out this specific reason for errors.

Q16. Answer: D) shapes that

Why? 'Shapes that often integrate mathematical curves' — 'that' starts a clause describing which shapes. No punctuation needed before 'that.' A period, colon, or semicolon would cut the sentence in half and leave a fragment.

Q17. Answer: D) following. In

Why? 'Has gained an international following' is a complete sentence. 'In 2018, the duo...' is another complete sentence. A period correctly separates them. A comma would create a run-on.

Q18. Answer: C) languages

Why? 'Including the languages Tamil and Kannada' — no punctuation needed between 'languages' and the specific names. The names directly identify which languages. A comma, colon, or dash would incorrectly break the phrase.

Q19. Answer: A) geology; the

Why? Two complete thoughts: 'they differ in geology' and 'Vesta has a basaltic surface...' A semicolon correctly connects two related complete sentences. It shows they are closely linked ideas.

Q20. Answer: C) offsets

Why? The subject is 'mechanism' (singular) describing a general scientific fact (present tense). 'Offsets' is present tense, singular — matches perfectly. 'Had offset' and 'offset' are past tense. 'Offsetting' can't be the main verb alone.

Q21. Answer: D) establish

Why? The subject is 'rates and trends' — joined by 'and,' making it plural. So the verb must be plural: 'establish.' The phrase 'not its population size' is just a side note and doesn't change the plural subject. 'Establishes' is singular — wrong.

Q22. Answer: D) next,

Why? The sentence says 'first examine...' so the next step needs a sequence word. 'Next' continues the sequence. 'Therefore' implies a conclusion. 'Still' implies contrast. 'Instead' implies replacement.

Q23. Answer: B) consequently,

Why? The plant absorbs cadmium from the soil (cause). As a result, cadmium levels in the soil drop (effect). 'Consequently' means 'as a result' — perfect cause-and-effect transition.

Q24. Answer: A

Why? Goal: give an example of a country using 'franc.' A says 'Rwanda uses the Rwandan franc' — that IS the example. B says what Rwanda doesn't use. C discusses word origins. D gives a general count without naming a country.

Q25. Answer: A

Why? The audience does NOT know what historical analysis is, so you must explain it. A defines it: 'historical analysis examines the social and political circumstances surrounding a work's creation.' Simple and clear for newcomers.

Q26. Answer: C

Why? Goal: show a structural DIFFERENCE. C uses 'Unlike' to directly contrast: graphene = flat hexagonal sheets vs. diamond = rigid 3D lattice. The word 'Unlike' makes the difference crystal clear. D states the same facts but without the contrast word.

Q27. Answer: C

Why? Goal: COMPARE the two frameworks. C says: 'Both address how organisms learn, but they disagree on whether observation alone can produce learning.' Skinner says you must do it yourself. Bandura says you can learn by watching others. That is the key difference.

SECTION 1: READING & WRITING — MODULE 2 HARDER (27 Questions)

Q1. Answer: B) static

Why? The passage says the marshes are NOT _____, then describes constant measurable changes. The word must mean 'unchanging.' 'Static' = not moving or changing. So 'not static' = always changing. Fits perfectly.

Q2. Answer: A) derivative

Why? The critic says the designs rely on techniques that have been 'standard for over a decade' and add 'little genuinely novel.' That means they lack originality. 'Derivative' = based on existing work, not original.

Q3. Answer: B) complication

Why? The varying expertise of volunteers forces scientists to create complex statistical fixes before using the data. That is a difficulty/problem. 'Complication' = something that makes things harder. 'Innovation' is positive (wrong). 'Redundancy' means repetition. 'Incentive' means motivation.

Q4. Answer: A

Why? Archer sees nothing wrong with society and views himself as its 'rightful custodian.' He is completely comfortable — not questioning anything. This sets up a novel that will LATER challenge those views. That is A.

Q5. Answer: A

Why? Saldana makes a claim. Mensah challenges it. Petrov brings both views together. Claim → counterargument → reconciliation = choice A.

Q6. Answer: B

Why? The narrator keeps coming back because understanding Sensei seems 'perpetually, tantalizingly close' — always almost within reach but never quite there. The mystery itself is the attraction. That is B.

Q7. Answer: B

Why? Hagen says abyssal plains block species from spreading between vents. Shankar found almost NO genetic differences across 4,000 km — meaning species DO successfully travel those distances. So Shankar would say: the species cross those 'barriers' better than Hagen thinks.

Q8. Answer: B

Why? The hypothesis: code-switching helps executive function. To CHALLENGE this, you need evidence it does NOT help. B says the most frequent code-switchers showed NO advantage at all — that directly contradicts the hypothesis.

Q9. Answer: B

Why? Prasad says volcanic eruptions (lasting 800,000 years) released the iridium, not a single asteroid hit. If volcanism did it gradually, you'd expect iridium spread across a WIDE time period (not one thin sharp layer). B describes exactly this: a broad interval, not a spike.

Q10. Answer: B

Why? The finding: more rainfall variability = more insurance claims. To best illustrate this, pick the pair with the BIGGEST spread in both measures. Province B (31% variability, 14.8% claims) and Province D (45%, 22.1%) show the largest differences, making the trend clearest.

Q11. Answer: A) 0.62 W/m·K at 150°C

Why? The table says Composite P: 0.45 at 25°C, 0.62 at 150°C. Just read the value directly from the table. The other choices are values for different composites.

Q12. Answer: A

Why? The claim: the marketplace is alive with the vendors' personal stories. A shows Farida's spice blends came 'Directly from the kitchen of her own memory' — not from any recipe. Her personal history is what makes the stall special.

Q13. Answer: B) volumes,

Why? The phrase 'characterized by sweeping curves and fragmented geometric volumes' is a descriptive aside set off by commas. You need a comma after 'volumes' to close this aside and let the main verb 'challenged' continue.

Q14. Answer: D) and would also

Why? 'MPAs would safeguard habitats and would also serve as a baseline.' This keeps the grammar parallel: 'would safeguard... and would also serve.' Clean and correct.

Q15. Answer: A) implying

Why? 'Certain genes have resisted transfer, implying that...' — 'implying' is a present participle that smoothly adds a consequence to the main clause. 'Implied' (past), 'to imply' (infinitive), and 'having implied' (completed action) don't fit the sentence flow.

Q16. Answer: B) were achievable

Why? 'Neither... nor...' — the verb agrees with the closer subject, which is 'arcs' (plural). So you need a plural verb: 'were.' Past tense is correct because the innovations happened in the mid-twentieth century.

Q17. Answer: B) is

Why? 'The genomic data' refers to one body of information (treated as singular here). 'Is' matches a singular subject in the present tense. The data collection is ongoing, so past tense options are wrong.

Q18. Answer: C) Moreover,

Why? The passage lists one effect of thawing (greenhouse gases released), then adds a SECOND effect (ground becomes unstable). 'Moreover' means 'in addition' — you're adding another related point.

Q19. Answer: D) However,

Why? First: optimism about superconductors. Then: engineering obstacles blocked progress. This is a shift from good news to bad news. 'However' signals contrast/reversal.

Q20. Answer: C) however,

Why? Urban heat islands are well known. Urban COOL islands are less known and somewhat surprising. 'However' introduces this unexpected twist — something that goes against what you might expect.

Q21. Answer: A

Why? Goal: practical applications. A says modifying crops 'could extend growing seasons in high-latitude regions.' That is a real-world use of the research. The other choices describe history, validation, or mechanisms — not applications.

Q22. Answer: B

Why? Goal: show a DIFFERENCE between Art Nouveau and Bauhaus. B uses 'Whereas' to contrast them: Art Nouveau = elaborate organic decoration vs. Bauhaus = plain geometric simplicity with industrial materials. Clear contrast.

Q23. Answer: D

Why? The audience ALREADY KNOWS what bioaccumulation is. So don't waste time defining it. D gives the specific finding (mussels at low-oxygen sites had 3x higher rates) without re-explaining the concept. B defines it unnecessarily.

Q24. Answer: B

Why? Goal: compare how the two poets approached form. B contrasts Okigbo's 'fragmented imagery from European modernism' with Soyinka's 'Yoruba oral literature traditions.' Two different formal approaches, directly compared.

Q25. Answer: A

Why? Goal: highlight a SURPRISE. A explicitly says the researchers 'expected' one outcome but found the opposite in old-growth forests. The word 'expected' followed by 'but' makes the surprise obvious.

Q26. Answer: A

Why? Goal: show the research was INTERDISCIPLINARY (combined multiple fields). A names both: 'satellite imagery from remote sensing' + 'radiocarbon dating from archaeology.' Two fields working together = interdisciplinary.

Q27. Answer: **C**

Why? Goal: emphasize the SIZE of the gap between prediction and measurement. C gives exact numbers (0.877 predicted vs. 0.831 measured), the percentage (5.2%), and calls it 'statistically significant.' Maximum emphasis on scale.

SECTION 2: MATH — MODULE 1 (22 Questions)

Q1. Answer: D) 276

Why? Width = 6. Length = 6 + 40 = 46. Area = length $\times$ width = $46 \times 6 = 276$ sq cm.

Q2. Answer: 60

Why? Start at 42 cm. Add 1.8 cm/day for 10 days: $42 + (1.8 \times 10) = 42 + 18 = 60$ cm.

Q3. Answer: C) $v = -1.2 + 2.0u$

Why? From the graph, the line crosses the v-axis around -1.2 (y-intercept) and goes UP with a slope of about 2. Only C has intercept -1.2 and positive slope $+2.0$.

Q4. Answer: 9

Why? Set equal: $-0.8 = x^2 + 6x + a \rightarrow x^2 + 6x + (a + 0.8) = 0$. For NO real solutions, the discriminant must be negative: $36 - 4(a + 0.8) < 0 \rightarrow a > 8.2$. The smallest whole number above 8.2 is 9.

Q5. Answer: B) (0, -3)

Why? The y-intercept is where $x = 0$. Plug in: $f(0) = 4(0) - 3 = -3$. So the y-intercept is the point (0, -3).

Q6. Answer: C) 52

Why? $7(x+3) = 6(x+3) + 52$. Subtract $6(x+3)$ from both sides: $1 \times (x+3) = 52$. So $x + 3 = 52$.

Q7. Answer: A

Why? In $25a + 18b = 845$, the number 25 is multiplied by a (hours on Project A). So 25 represents the dollars earned PER HOUR on Project A.

Q8. Answer: C) $y = 51.6$

Why? The question asks what y (total grain consumed) equals. The goose ate 51.6 pounds total. So $y = 51.6$. Simple.

Q9. Answer: 4.40

Why? Radius = $2/2 = 1$ cm. Volume = $\pi \times 1^2 \times 4 = 4\pi \approx 12.566$ cm³. Mass = density $\times$ volume = $0.350 \times 12.566 \approx 4.40$ grams.

Q10. Answer: 13

Why? Combine like terms: $5x^{\blacksquare} + 11x^{\blacksquare} - 3x^{\blacksquare} = (5 + 11 - 3)x^{\blacksquare} = 13x^{\blacksquare}$. So $b = 13$.

Q11. Answer: C) $\sqrt{(246y)}$

Why? $(246y)^{(1/2)}$ means 'the square root of 246y' which is written as $\sqrt{(246y)}$. That is choice C. Choice A separates the 246 from the square root. Choice D squares instead of square-rooting.

Q12. Answer: B) 48

Why? $P(\text{room A}) + P(\text{room B}) + P(\text{room C}) = 1$. So $P(C) = 1 - 0.55 - 0.30 = 0.15$. Number in C = $0.15 \times 320 = 48$.

Q13. Answer: B) 3

Why? Line p has slope $-1/3$. Perpendicular lines have slopes that are negative reciprocals. The negative reciprocal of $-1/3$ is $+3$.

Q14. Answer: A

Why? Plug in each x-value: $f(-4) = -8 \times 3^{(-1)} = -8/3$ ✓. $f(0) = -8 \times 1 = -8$ ✓. $f(4) = -8 \times 3 = -24$ ✓. $f(8) = -8 \times 9 = -72$ ✓. All match choice A.

Q15. Answer: B) $12x + 96$

Why? Distribute: $12(x + 8) = 12 \times x + 12 \times 8 = 12x + 96$.

Q16. Answer: 31

Why? $g(4) = 3(4) + 5 = 17$. $h(4) = 7 - 4 = 3$. Then $2g(4) - h(4) = 2(17) - 3 = 34 - 3 = 31$.

Q17. Answer: 215

Why? Each laptop with 6% tax costs $210 \times 1.06 = \$222.60$. Divide budget: $48,000 \div 222.60 = 215.6$. You can only buy whole laptops, so maximum = 215.

Q18. Answer: A

Why? To convert radians to degrees, multiply by $180^\circ/\pi$. So $-(7/10)\pi \times (180^\circ/\pi)$ gives the answer in degrees. Choice B uses $\pi/180^\circ$ (that converts degrees TO radians — backwards). Choice C has a positive sign (missing the negative). Choice D uses 360° instead of 180° .

Q19. Answer: D) $M = 200(0.82)^{(t/31.5)}$

Why? Loses 18% means 82% REMAINS each cycle. The retention factor is 0.82 (not 0.18). The cycle is 31.5 minutes, so exponent = $t/31.5$. Start with 200 grams: $M = 200(0.82)^{(t/31.5)}$.

Q20. Answer: 102

Why? Find y : $42 + y + (3y - 6) = 180 \rightarrow 4y = 144 \rightarrow y = 36$. Angle $F = 3(36) - 6 = 102^\circ$. Since GH is parallel to DF , angle $EHG =$ angle $F = 102^\circ$ (corresponding angles).

Q21. Answer: 320/3

Why? Perimeter = 960, so each side = 320. The circumradius of an equilateral triangle = $\text{side} \times \sqrt{3} / 3 = 320\sqrt{3}/3$. Since $R = k\sqrt{3}$, we get $k = 320/3$.

Q22. Answer: B) 322

Why? Vertex (3, 10): $f(x) = a(x-3)^2 + 10$. Using point (4, 56): $a(1) + 10 = 56 \rightarrow a = 46$. Then $f(-1) = 46(16) + 10 = 746$. $f(0) = 46(9) + 10 = 424$. Answer: $746 - 424 = 322$.

SECTION 2: MATH — MODULE 2 EASIER (22 Questions)

Q1. Answer: A

Why? In $24b + 18p = 936$, the number 24 is multiplied by b (bags of bread flour). It represents the cost per bag of bread flour: \$24.

Q2. Answer: 6312.5

Why? Birth weight = 6200 lb. Gains 12.5 lb/hr for 9 hours: $6200 + (12.5 \times 9) = 6200 + 112.5 = 6312.5$ pounds.

Q3. Answer: B) $q = s + 28r$

Why? Start with $q - 28r = s$. Add $28r$ to both sides: $q = s + 28r$. Done.

Q4. Answer: D) (5, -2)

Why? Test each point against the inequality shown in the graph. The point (5, -2) falls inside the shaded region. The other three points fall outside it.

Q5. Answer: A) The length of the rectangle, in meters

Why? The rectangle has width = w and length = $36w$. In the equation $y = (36w)(w)$, the factor $36w$ is the length.

Q6. Answer: C) 41

Why? $8(x+5) = 7(x+5) + 41$. Subtract $7(x+5)$ from both sides: $(x+5) = 41$.

Q7. Answer: B) 11

Why? Rewrite the equations: $x + 7y = -6$ and $x - 7y = 17$. Add them together: $2x = 11$. The question asks for $2x$, which is 11.

Q8. Answer: C) $y = 42.5$

Why? The total amount of salmon consumed in 10 days was 42.5 pounds. So $y = 42.5$.

Q9. Answer: 22.5

Why? Volume of cube = $5^3 = 125 \text{ cm}^3$. Mass = density $\times$ volume = $0.180 \times 125 = 22.5$ grams.

Q10. Answer: B) 0 and -4

Why? Simplify: $-3|4x + 8| = -24 \rightarrow |4x + 8| = 8$. Two cases: $4x + 8 = 8$ gives $x = 0$. $4x + 8 = -8$ gives $x = -4$. Both solutions.

Q11. Answer: B) 6

Why? From p : $p(x) = 6x + 14$, so $p(d) = 6d + 14 = -4$, giving $d = -3$. For t : $t(-3) = -4$ and $t(7) = 56$. Slope of $t = (56 - (-4)) / (7 - (-3)) = 60/10 = 6$.

Q12. Answer: B) $Kx - 10y = -24$

Why? Rearrange B: $y = (K/10)x + 2.4$. This gives positive slope (since K is positive), y -intercept of 2.4, and negative x -intercept — all matching the graph.

Q13. Answer: C) -2

Why? From the graph's boundary line and shading direction, determine r and t in the equation $rx + ty = -48$. The intercepts and slope give $r + t = -2$.

Q14. Answer: A) 25/5,184

Why? Edge of cube B = $(5/72) \times$ edge of cube A. Surface area scales as the SQUARE of the linear scale factor: $n = (5/72)^2 = 25/5,184$.

Q15. Answer: B) $f(x) = -5^x + 4$

Why? The graph shows $y = f(x) + 4$. Using a key point from the graph to find $f(0)$, then solving for the constant C in $f(x) = -5^x + C$ gives $C = 4$.

Q16. Answer: B) $18x + 126$

Why? Distribute: $18(x + 7) = 18x + 18(7) = 18x + 126$.

Q17. Answer: D) 37.50

Why? Original map: 28 in. per side, 1 in. = 15 mi. Total distance across = $28 \times 15 = 420$ mi. Smaller map side = $28 \times 0.40 = 11.2$ in. (60% shorter). Same 420 mi on 11.2 in.: $420 \div 11.2 = 37.5$ mi per inch.

Q18. Answer: C) $241/16$

Why? Substitute and simplify: $4x^2 - 79x + (400 - d) = 0$. Two distinct solutions need discriminant > 0 : $d > 159/16 = 9.9375$. Only C (= 15.0625) is greater than 9.9375. A (5), B (9), and D (8) are all too small.

Q19. Answer: C

Why? We already know angle G = angle J = 62° . Choice C tells us angle H = 44° and angle K = 74° . Then triangle GHI has angles {62, 44, 74} and triangle JKL has angles {62, 74, 44} — same set of angles. AA similarity confirmed.

Q20. Answer: 80

Why? Perimeter = 720 $\rightarrow$ side = 240. Circumradius $R = 240\sqrt{3}/3 = 80\sqrt{3}$. Since $R = m\sqrt{3}$, $m = 80$.

Q21. Answer: 84

Why? $f(14a) = 14a/a - 20 = 14 - 20 = -6$. $f(6a) = 6a/a - 20 = 6 - 20 = -14$. Product = $(-6)(-14) = 84$.

Q22. Answer: 6250

Why? $f(6)/f(3) = d^{(3/m)} = 250/10 = 25$. From $f(3)$: $c \times 25 = 10 \rightarrow c = 2/5$. Then $f(9) = c \times 25^3 = (2/5) \times 15,625 = 6,250$.

SECTION 2: MATH — MODULE 2 HARDER (22 Questions)

Q1. Answer: C) 160

Why? Let flat fee = F , hourly rate = r . Two equations: $F + 3r = 340$ and $F + 7r = 580$. Subtract: $4r = 240 \rightarrow r = 60$. Then $F = 340 - 3(60) = 160$.

Q2. Answer: B) 8

Why? Add the two equations: $(3x + 5y) + (7x - 5y) = 26 + (-6) \rightarrow 10x = 20 \rightarrow x = 2$. So $4x = 4(2) = 8$.

Q3. Answer: A) $x = 4$ and $x = 10$

Why? Factor: $x^2 - 14x + 40 = (x - 4)(x - 10) = 0$. Check: $-4 \times -10 = 40$ ✓ and $-4 + -10 = -14$ ✓. Solutions: $x = 4$ and $x = 10$.

Q4. Answer: B) 4

Why? Divide: $(3x^2 + 14x - 5) \div (x + 5)$. Factor the top: $3x^2 + 14x - 5 = (x + 5)(3x - 1)$. So the result is $3x - 1$, meaning $a = 3$, $b = -1$. Then $a - b = 3 - (-1) = 4$.

Q5. Answer: C) 100

Why? When you add the same number to every value, the mean increases by that number. Old mean = 85. Add 15 to each $\rightarrow$ new mean = $85 + 15 = 100$.

Q6. Answer: B) 4.5

Why? The x -intercepts are where each factor = 0: $x = 7$, $x = -3$, $x = 1/2$. Sum = $7 + (-3) + 0.5 = 4.5$.

Q7. Answer: B) 16

Why? $(x - 4)^2 + (y + 9)^2 = 64$ is a circle. The right side is $r^2 = 64$, so radius = 8. Diameter = $2 \times 8 = 16$.

Q8. Answer: B) 74.5

Why? Use the line equation: score = slope $\times$ hours + intercept = $6.5(5) + 42 = 32.5 + 42 = 74.5$.

Q9. Answer: B) $P = 500(2)^{t/4.5}$

Why? Doubles = growth factor of 2. Doubling period = 4.5 hours. So exponent = $t/4.5$. Starting population 500: $P = 500 \times 2^{t/4.5}$.

Q10. Answer: B) 0 and -6

Why? Simplify: $-5|3x + 9| = -45 \rightarrow |3x + 9| = 9$. Case 1: $3x + 9 = 9 \rightarrow x = 0$. Case 2: $3x + 9 = -9 \rightarrow x = -6$.

Q11. Answer: C) 5

Why? Step by step: $g(4) = 16 + 1 = 17$. $f(17) = 2(17) - 3 = 31$. Then $f(4) = 2(4) - 3 = 5$. $g(5) = 25 + 1 = 26$. Finally: $31 - 26 = 5$.

Q12. Answer: A) 5/13

Why? In a right triangle, if the right angle is at Q , then angles P and R add up to 90° . This means $\cos(R) = \sin(P)$. Since $\sin(P) = 5/13$, $\cos(R) = 5/13$.

Q13. Answer: C) \$96

Why? Selling price = cost $\times 1.25$ (a 25% markup). So $120 = 1.25 \times \text{cost} \rightarrow \text{cost} = 120 \div 1.25 = \96 .

Q14. Answer: C) 18

Why? $h(x) = -2(x - 5)^2 + 18$ is a downward parabola. The vertex is at $(5, 18)$. The highest point on a downward parabola IS the vertex. Maximum value = 18.

Q15. Answer: C) 9

Why? Cross-multiply: $2x + 6 = 4(x - 3) = 4x - 12$. Rearrange: $18 = 2x \rightarrow x = 9$. Check: $(18 + 6)/(9 - 3) = 24/6 = 4$ ✓.

Q16. Answer: C) 11/50

Why? Total = 25 marbles. $P(\text{1st blue}) = 12/25$. Without replacement, $P(\text{2nd blue}) = 11/24$. Multiply: $(12/25)(11/24) = 132/600 = 11/50$. Choice A uses replacement (wrong). Choice D uses 25 in the second draw (wrong).

Q17. Answer: 15

Why? Volume = $\pi r^2 h$. So $\pi r^2 \times 6 = 1350\pi \rightarrow r^2 = 1350/6 = 225 \rightarrow r = \sqrt{225} = 15$ cm.

Q18. Answer: -47/3

Why? Substitute $y = 2x + k$ into $y = -3(x-8)^2$: $3x^2 - 46x + (k+192) = 0$. Exactly one solution means discriminant = 0: $46^2 - 12(k+192) = 0 \rightarrow 2116 - 12k - 2304 = 0 \rightarrow k = -188/12 = -47/3$.

Q19. Answer: B) 2

Why? Remainder Theorem: divide by $(x - 2) \rightarrow$ remainder = $f(2)$. Calculate: $4(8) - 7(4) + 5(2) - 12 = 32 - 28 + 10 - 12 = 2$.

Q20. Answer: 13.5 (or 27/2)

Why? Slope of line $m = (17-5)/(8-2) = 2$. Perpendicular slope = $-1/2$. Midpoint = $(5, 11)$. Line n : $y - 11 = -1/2(x - 5) \rightarrow y = -x/2 + 27/2$. At $x = 0$: $y = 27/2 = 13.5$.

Q21. Answer: 2

Why? $f(x) = a(x-3)^2 - 4$, vertex at $(3, -4)$. Plug in $(0, 14)$: $a(9) - 4 = 14 \rightarrow 9a = 18 \rightarrow a = 2$.

Q22. Answer: 972

Why? $g(8)/g(4) = d^{(4/m)} = 108/12 = 9$. Then $g(12) = g(8) \times 9 = 108 \times 9 = 972$. Each step multiplies by the same factor of 9.

END OF ANSWER KEY — 147 QUESTIONS TOTAL

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